PANGAMBAM LENIN SINGH

me25mtech14005@iith.ac.in | leninpsingh16@gmail.com | +91 9863583938 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Kendriya Vidyalaya No.1, Lamphelpat, Imphal	Class XII (CBSE)	2021	91.00%
Kendriya Vidyalaya No.1, Lamphelpat, Imphal	Class X (CBSE)	2019	90.80%

SKILLS

(1) Python fundamentals

https://acrobat.adobe.com/id/urn:aaid:sc:AP:a11dce56-ddfb-4563-b7ce-a5ca0ff42b14

(2) Numerical analysis through Python

https://acrobat.adobe.com/id/urn:aaid:sc:AP:2f3b6d59-b633-4fb8-83d4-cc2bf48876b8

PROJECTS

(1) M.Tech. thesis on 'Python Modelling of Single Aluminum Particle Combustion'

- Replication of 'Detailed Modeling of Aluminum Particle Combustion - From Single Particles to Cloud Combustion in Bunsen Flames' by Zhang et. al.
- **Lagrangian particle sub-model** for the particle-phase processes
- **Eulerian gas sub-model** for the gas-phase processes
- **Self-test suite** for mass, energy and atom conservation
- Extension to variation of **domain cell resolution**
- Incorporation of variation of **inflow composition**

(2) B.Tech. project on 'Aerodynamic Analysis of a Delta Wing'

- Geometry modelling of a cropped delta wing in **SOLIDWORKS**
- **3D subsonic simulation** of a cropped delta wing in **Ansys Fluent**
- k-omega SST turbulence model
- Parametric study of sweep angle, taper ratio and semi-span
- Best overall aerodynamic performance from the baseline wing configuration
- Extension to high angles of attack
- Local mesh refinement and transient simulations for non-converging, unsteady flow
- Results of stall angle and maximum lift coefficient
- Comparison against **NACA 2412 and 4412 airfoils**
- Extension to **supersonic simulation**
- Density-based compressible solver with the k-epsilon realizable model
- Comparison of the lift and drag trends with the subsonic results

https://acrobat.adobe.com/id/urn:aaid:sc:AP:2bae7043-cf20-4b9a-9878-1e49c53bb732

(3) Course term project on 'Optimization of Divergent Angle of a Rocket Engine Nozzle Using CFD'

- Replication of the Kuttan and Sajesh's IJES research paper
- Compressible **supersonic flow** through a **2D conical de Laval nozzle** in **Ansys Fluent**
- Density-based transient solver with the standard k-epsilon turbulence model
- **Ansys Workbench** for geometry modelling, meshing and simulation
- DesignModeler, Meshing, and Fluent
- Variation of the divergent half-angle
- Divergent section analysis

- Mach number, static pressure, and turbulent intensity variation
- Diamond patterns of **oblique shock**
- Larger divergent angles weaken the **shock pattern** and raise the **exit Mach number**

https://acrobat.adobe.com/id/urn:aaid:sc:AP:0439142a-117d-44bd-b82e-7e9a9ea11d57

https://acrobat.adobe.com/id/urn:aaid:sc:AP:b0dcdf6d-d07b-4ca6-9e7c-31b368eace4f